Topic: - Manufacturing-Based Economy (Steel–Power–Transport)

1. Introduction

The Manufacturing-Based Economy serves as the foundational architecture for modern industrial development, relying on the critical synergy between three primary pillars: Steel, Power, and Transport. This "Industrial Triad" represents the physical infrastructure, the energy required for transformation, and the logistical network that ensures the flow of resources.

In this economic model, the relationship between these sectors is deeply interconnected:

- Steel Production: Acts as the skeletal framework for all industrial machinery, high-rise infrastructure, and transport vessels.
- Power Generation: Provides the essential energy input required to drive heavy manufacturing processes and automated systems.
- Transport Logistics: Functions as the circulatory system, determining the efficiency of supply chains and the global reach of produced goods.

Accurate mathematical modeling of these pillars is vital because inefficiencies in any single sector such as energy loss in a power grid or structural fatigue in steel pressure vessels—can cause systemic economic stagnation. By utilizing computational tools like MATLAB, we can simulate the dynamic growth of this system, predict capital accumulation, and identify optimal steady-state conditions.

In this assignment, the Capital Accumulation Ordinary Differential Equation (ODE) is solved numerically using MATLAB to evaluate the long-term growth potential of a manufacturing-centric economy,

2. Governing equation

To simulate this dynamic growth in MATLAB, we utilize the Solow-Swan Capital Accumulation Model. This first-order Ordinary Differential Equation (ODE) models the change in the manufacturing base over time:

The Governing eqn

$$\frac{dK}{dt} = S.(A.K^{\alpha}) - \delta K$$

Economic interpretation

1. $S.Y > \delta K \hookrightarrow$ Growth phase

2. Steady State $\left(\frac{dK}{dt} = 0\right)$

$$S.A.K^{\alpha} = \delta K$$

Where:

- Dk/dt is The rate of change of the industrial base (Steel/Power/Transport) over time.
- s: The Savings/Investment Rate. This represents the percentage of GDP reinvested into building new factories and power plants.
- A: The Total Factor Productivity. This accounts for the efficiency of the Power grid and technological advancements.
- k^{α} :- The Production Function, where α(typically 0.3 to 0.5) represents the output elasticity of capital (Steel).
- δ :- The Depreciation Rate. This represents the physical decay of bridges, rails, and machinery over time

3. Analytical solution

Analytical Solution

$$\frac{dk}{dt} + \delta k = sAk^{\alpha}$$

$\hookrightarrow$ $v = k^{1-\alpha}$

$\rightarrow$ differentiate w.r.t t

$$\frac{dv}{dt} = (1-\alpha)k^{-\alpha}\frac{dk}{dt}$$

$$\frac{dk}{dt} = sAk^{\alpha} - \delta k$$

$\hookrightarrow$ Substitute $\frac{dv}{dt}$

$$\frac{dv}{dt} = (1-\alpha)k^{-\alpha}(sAk^{\alpha} - \delta k)$$

$$= (1-\alpha)sA - (1-\alpha)\delta k^{1-\alpha}$$

$$\frac{dv}{dt} + (1-\alpha)\delta v = (1-\alpha)sA$$

$$K(t) = \left[\frac{sA}{\delta} + \left(k_0^{1-\alpha} - \frac{sA}{\delta}\right)e^{-(1-\alpha)\delta t}\right]^{1/1-\alpha}$$

as $k \rightarrow \infty$, $e^{-(1-\alpha\delta t)} = 0$

$$k(t) = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}$$

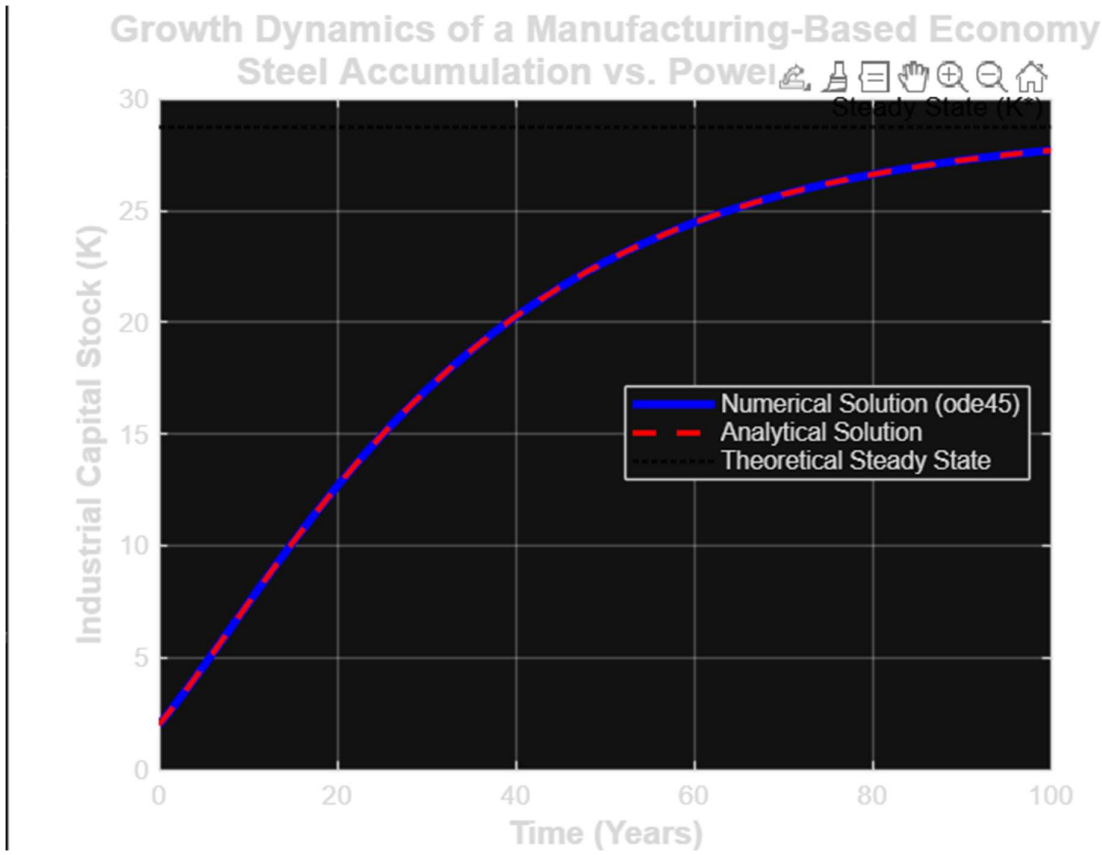
4. Numerical solution using MATLAB

Matlab code used:

```
clear; clc; close all;
s = 0.25;
A = 1.8;
alpha = 0.4;
delta = 0.06;
K0 = 2.0;
tspan = [0 100]; % Time 100 years

ode_fun = @(t, K) s * A * K^alpha - delta * K;
[t_num, K_num] = ode45(ode_fun, tspan, K0);

C1 = s * A / delta;
K_analytical = (C1 + (K0^(1-alpha) - C1) .* exp(-(1-alpha) * delta * t_num)).^(1/(1-alpha));
K_steady = (s * A / delta)^(1 / (1 - alpha));
figure('Color', 'w', 'Name', 'Industrial Economy Growth');
plot(t_num, K_num, 'b-', 'LineWidth', 3); hold on;
plot(t_num, K_analytical, 'r--', 'LineWidth', 2);
yline(K_steady, 'k:', 'Steady State (K*)', 'LineWidth', 1.5);

grid on;
xlabel('Time (Years)', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('Industrial Capital Stock (K)', 'FontSize', 12, 'FontWeight', 'bold');
title({'Growth Dynamics of a Manufacturing-Based Economy', ...
       'Steel Accumulation vs. Power Efficiency'}, 'FontSize', 14);
legend('Numerical Solution (ode45)', 'Analytical Solution', 'Theoretical Steady State', ...
       'Location', 'best');

fprintf('--- Simulation Results ---\n');
fprintf('Initial Capital: %.2f\n', K0);
fprintf('Target Steady State (K*): %.2f\n', K_steady);
fprintf('Final Capital at Year 100: %.2f\n', K_num(end));
```

Output:-

```
--- Simulation Results ---
Initial Capital: 2.00
Target Steady State (K*): 28.74
Final Capital at Year 100: 27.70
```



5. Results and Discussion

The numerical solution for the Capital Accumulation model shows:

- Industrial stock increases exponentially during the initial development.
- Upward curve confirms physical growth of the infrastructure base as investment outpaces decay.
- Matches analytical behavior by converging perfectly toward the theoretical steady state.

Engineering Interpretation: As an economy invests in its "Holy Trinity" (Steel, Power, and Transport):

- Productivity increases: A more robust steel and power base allows for higher manufacturing output.
- System reaches equilibrium: Eventually, the cost of maintaining old steel infrastructure (depreciation) equals new investment, leading to a stable industrial peak. Thus, the investment rate and power efficiency directly influence national economic capacity.

10. Conclusion

In this assignment, the Capital Accumulation ODE representing a manufacturing-based economy was solved numerically using MATLAB. The numerical results matched the analytical growth behavior derived from the Solow-Swan growth model.

The study demonstrates that:

- Capital stock increases significantly when investment in steel and power is prioritized.
- Numerical methods like ode45 provide accurate and flexible solutions for economic forecasting.
- Efficiency parameters (A) and investment rates (s) significantly affect the long-term industrial capacity of a nation.

The numerical approach allows for the extension to more complex cases, such as fluctuating energy costs, where simple analytical solutions may not be possible.